chapter 5

interaction design basics

interaction design basics

- design:
  - what it is, interventions, goals, constraints
- the design process
  - what happens when
- users
  - who they are, what they are like …
- scenarios
  - rich stories of design
- navigation
  - finding your way around a system
- iteration and prototypes
  - never get it right first time!

what is design?

achieving goals within constraints

- goals - purpose
  - who is it for, why do they want it
- constraints
  - materials, platforms, devices, time, costs, background.

golden rule of design

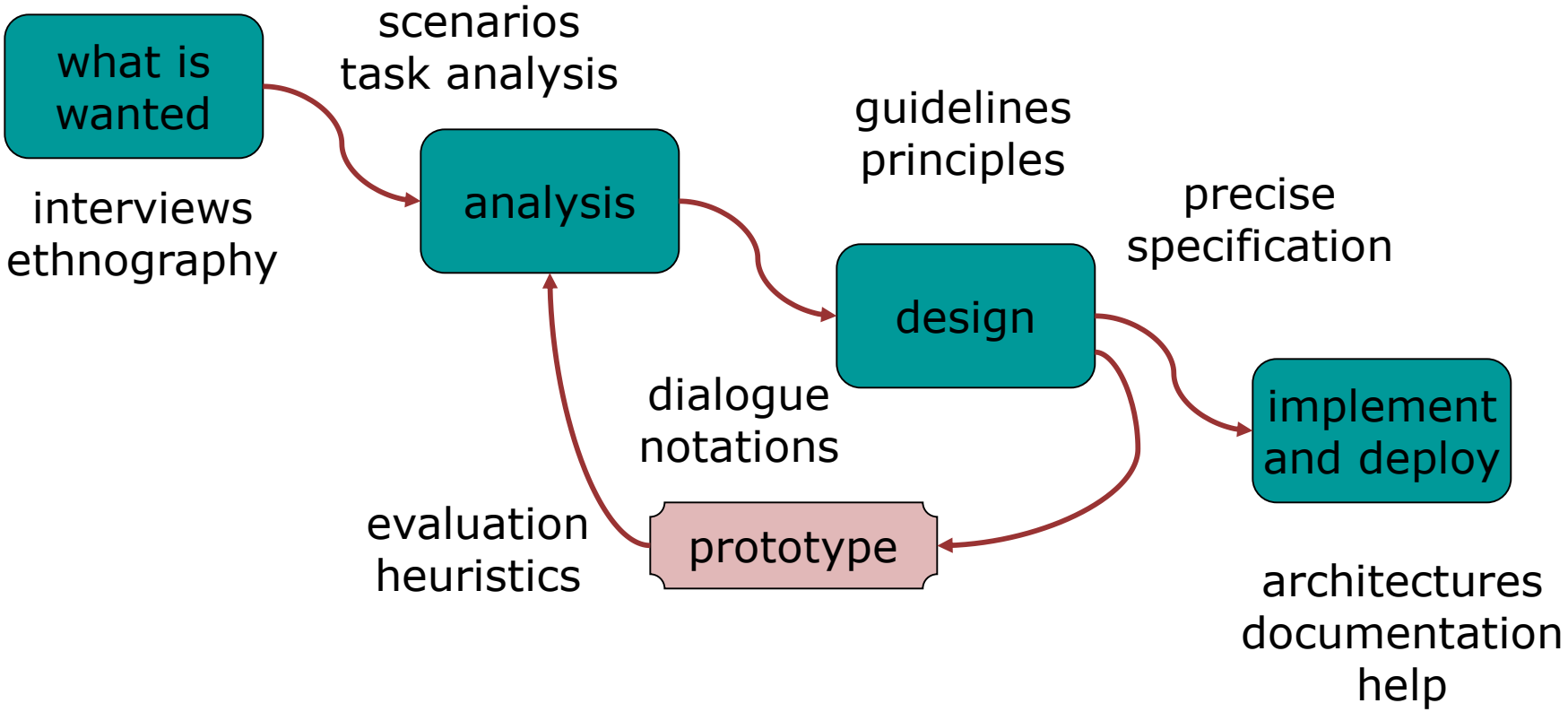
understand your materials

for Human-Computer Interaction

understand your materials

- understand computers
  - limitations, capacities, tools, platforms
- understand people
  - psychological, social aspects
  - human error
- and their interaction …

The process of design



Steps …

- requirements
  - what is there and what is wanted …
- analysis
  - ordering and understanding
- design
  - what to do and how to decide
- iteration and prototyping
  - getting it right … and finding what is really needed!
- implementation and deployment
  - making it and getting it out there

user focus

know your user

personae

cultural probes

know your user

- who are they?
- probably not like you!
- talk to them
- watch them
- use your imagination

persona

- description of an ‘example’ user
  - not necessarily a real person
- use as surrogate user
  - what would Betty think
- details matter
  - makes her ‘real’

scenarios

stories for design
use and reuse

scenarios

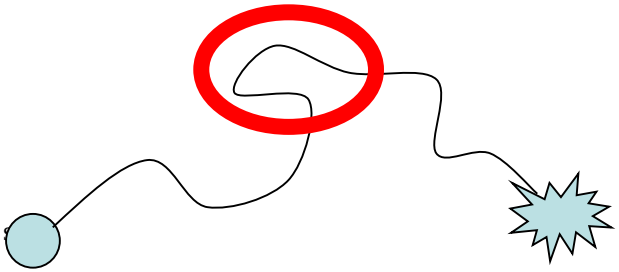
- stories for design
  - communicate with others
  - validate other models
  - understand dynamics
- linearity
  - time is linear - our lives are linear
  - but don’t show alternatives

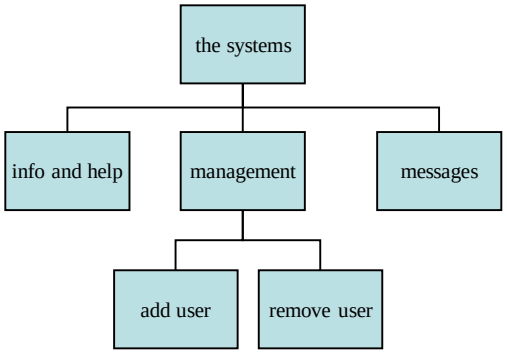
scenarios …

- what will users want to do?
- **step-by-step walkthrough**
  - what can they see (sketches, screen shots)
  - what do they do (keyboard, mouse etc.)
  - what are they thinking?
- use and reuse throughout design

use scenarios to ..

- communicate with others
  - designers, clients, users
- validate other models
  - 'play' it against other models
- express dynamics
  - screenshots – appearance
  - scenario – behaviour

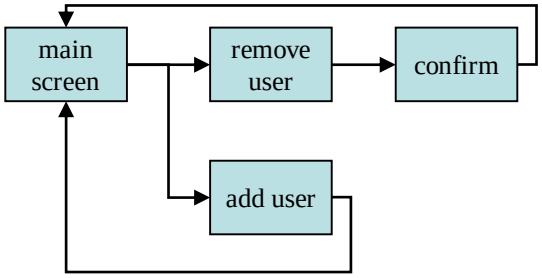




navigation design

local structure – single screen

global structure – whole site



levels

- widget choice
  - menus, buttons etc.
- screen design(arrangement buttons)
- application navigation design
- environment
  - other apps, O/S

think about structure

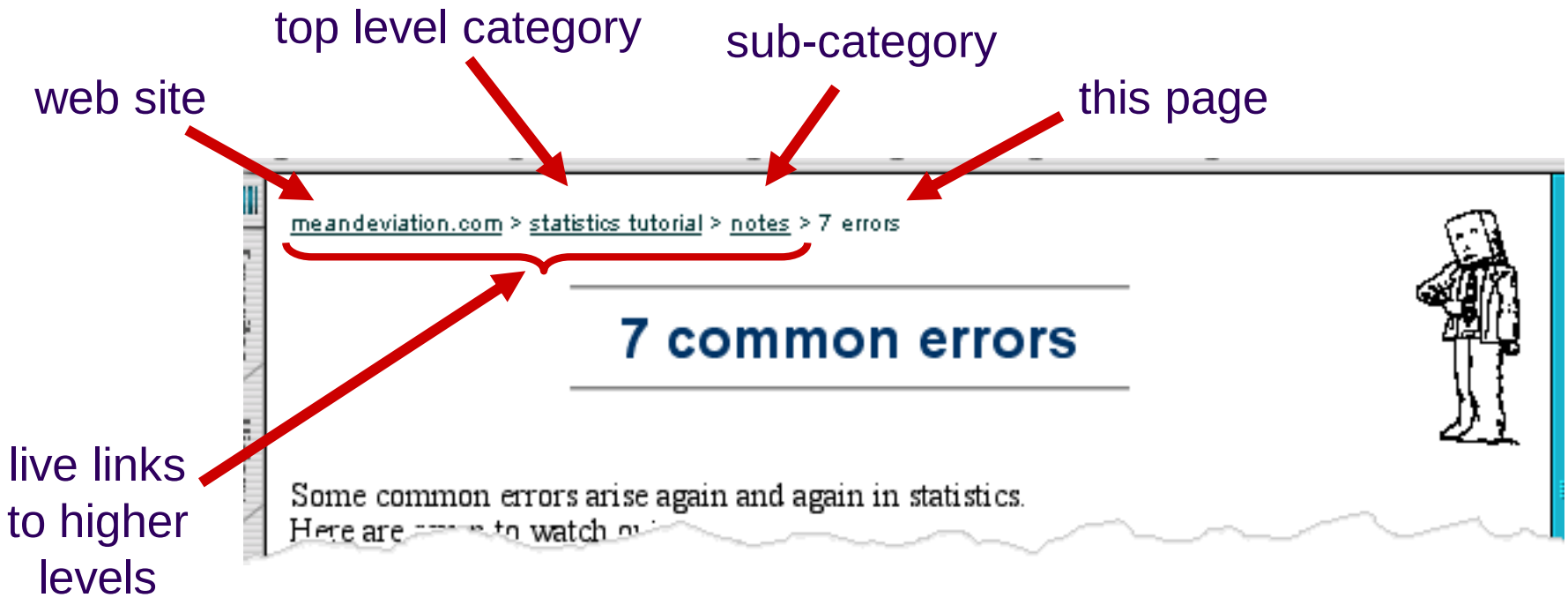
- within a screen
  - later ...
- local
  - looking from this screen out
- global
  - structure of site, movement between screens
- wider still
  - relationship with other applications

local

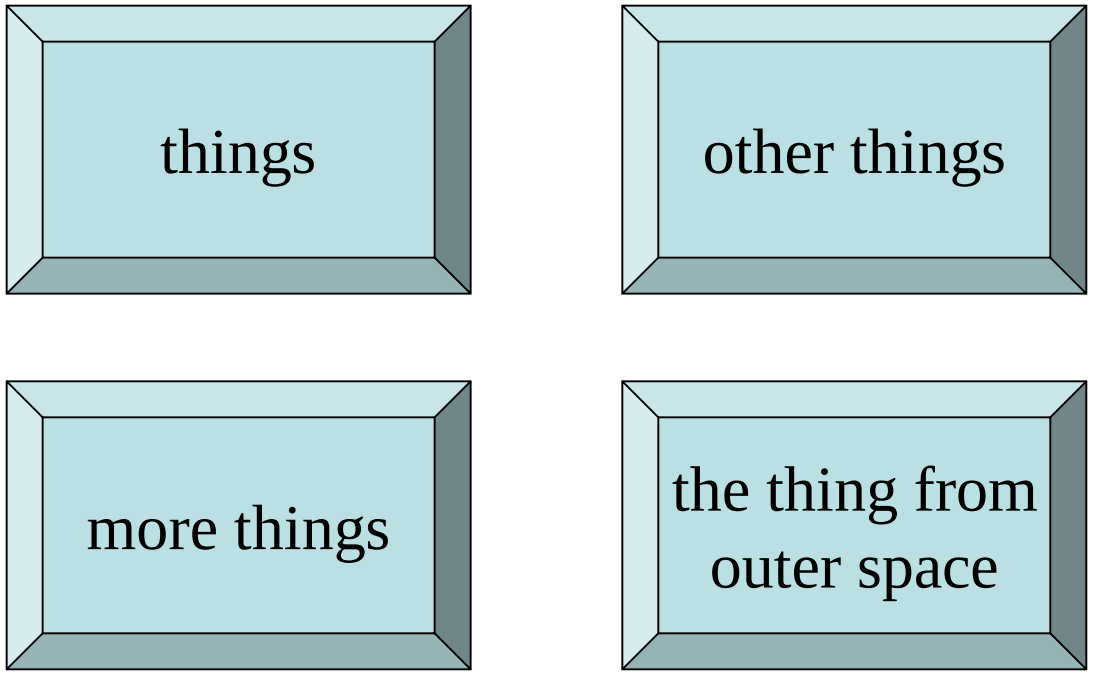
from one screen looking out

where you are – breadcrumbs

shows path through web site hierarchy



beware the big button trap

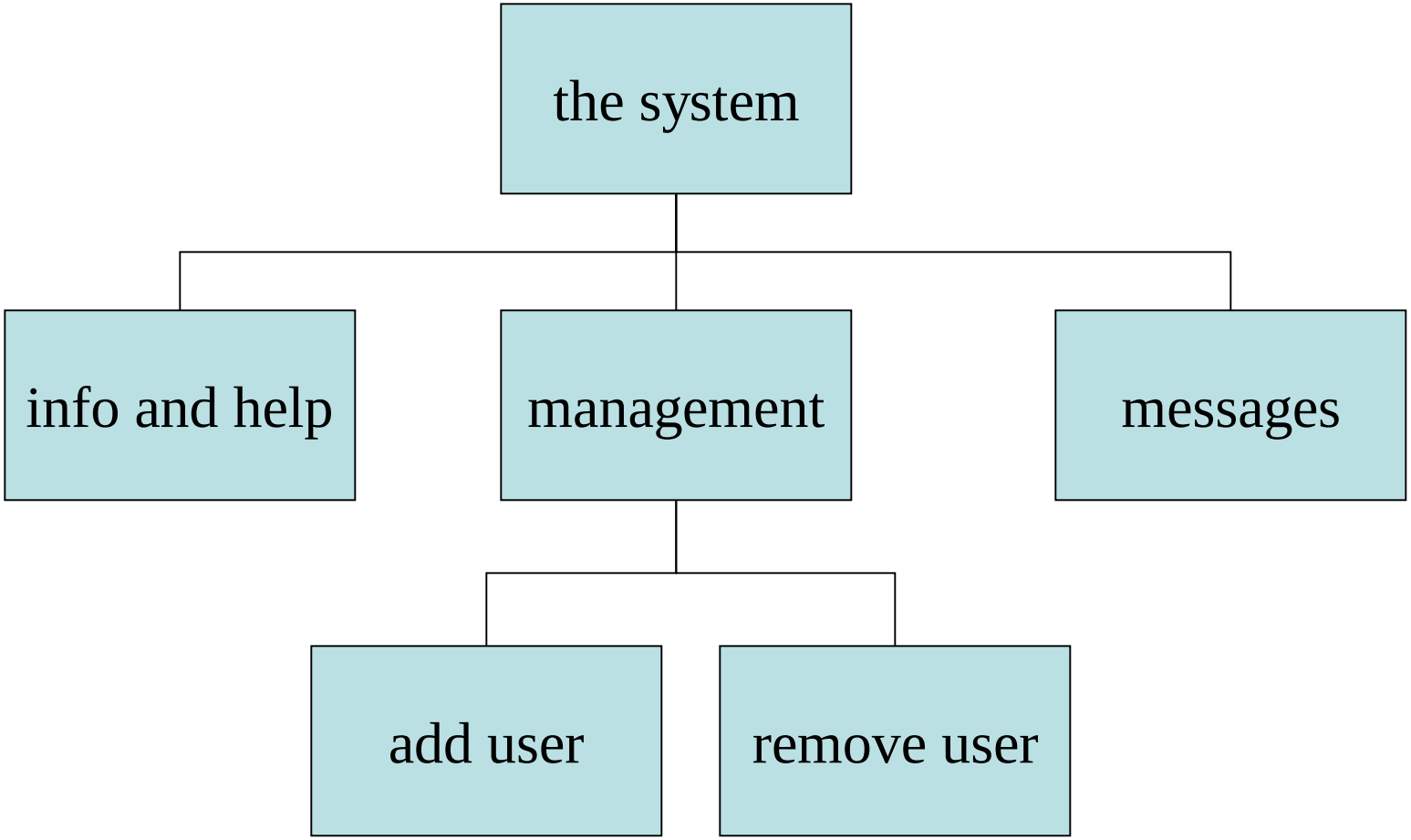


- where do they go?
  - lots of room for extra text!

Global structure

between screens
within the application

hierarchical diagrams

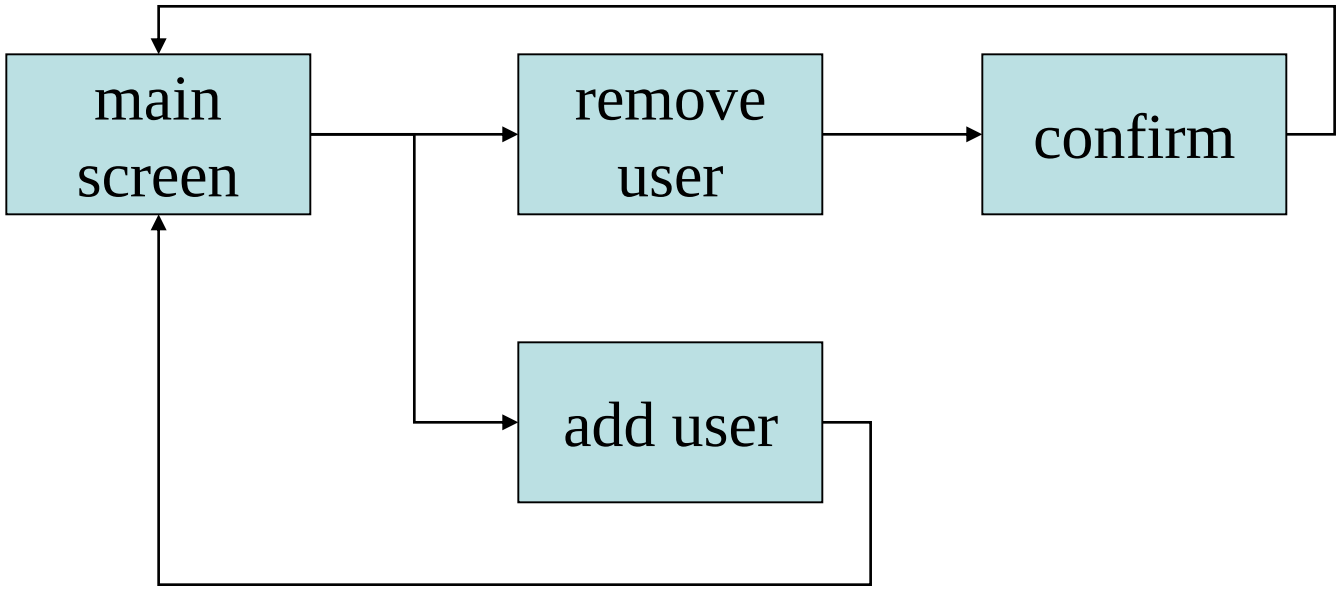


think about dialogue

what does it mean in UI design?

Minister: do you *name* take this woman …
Man: I do
Minister: do you *name* take this man …
Woman: I do
Minister: I now pronounce you man and wife

network diagrams



- show different paths through system

wider still …

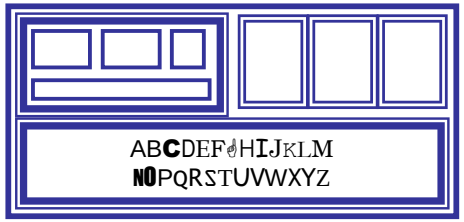
- style issues:
  - platform standards, consistency
- functional issues
  - cut and paste
- navigation issues
  - embedded applications
  - links to other apps … the web

Dix , Alan
Finlay, Janet
Abowd, Gregory
Beale, Russell



screen design and layout

basic principles
grouping, structure, order
alignment
use of white space



basic principles

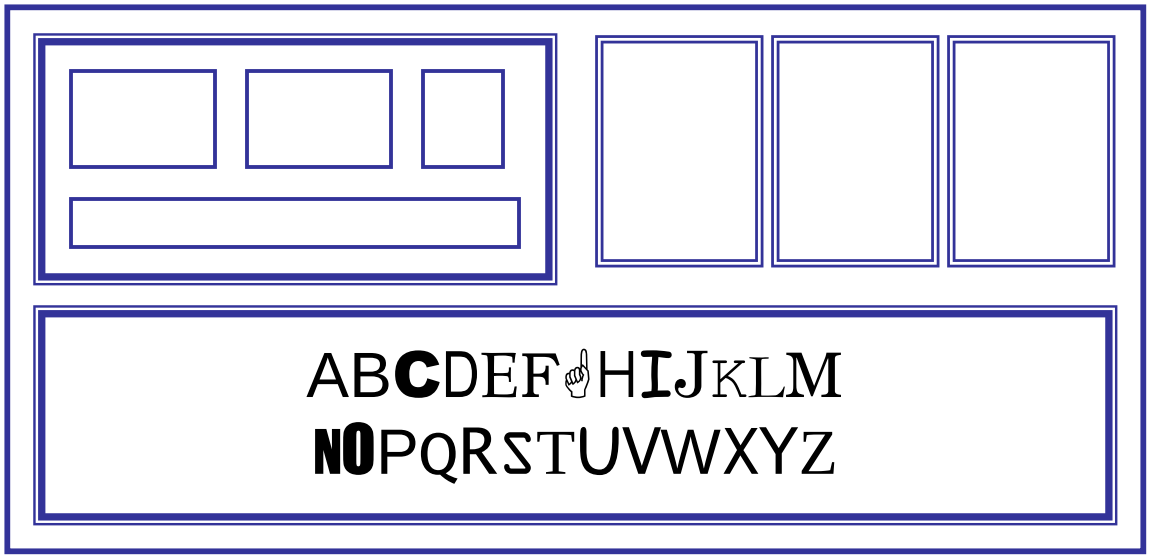
- ask
  - what is the user doing?
- think
  - what information, comparisons, order
- design
  - form follows function

available tools

- grouping of items
- order of items
- decoration - fonts, boxes etc.
- alignment of items
- white space between items

decoration

- use boxes to group logical items
- use fonts for emphasis, headings
- but not too many!!

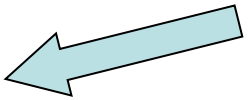


alignment - text

- you read from left to right (English and European)

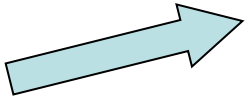
⇒ align left hand side

Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess



boring but readable!

fine for special effects but hard to scan



Willy Wonka and the Chocolate Factory
Winston Churchill - A Biography
Wizard of Oz
Xena - Warrior Princess

alignment - names

- Usually scanning for surnames
  ⇒ make it easy!

Alan Dix
Janet Finlay
Gregory Abowd
Russell Beale



Alan Dix
Janet Finlay
Gregory Abowd
Russell Beale



Dix , Alan
Finlay, Janet
Abowd, Gregory
Beale, Russell



alignment - numbers

think purpose!

which is biggest?

532.56
179.30
256.317
15
73.948
1035
3.142
497.6256

multiple columns

- scanning across gaps hard:

(often hard to avoid with large data base fields)

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

multiple columns - 2

- use leaders

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

multiple columns - 3

- or greying (vertical too)

sherbert	75
toffee	120
chocolate	35
fruit gums	27
coconut dreams	85

white space - the counter

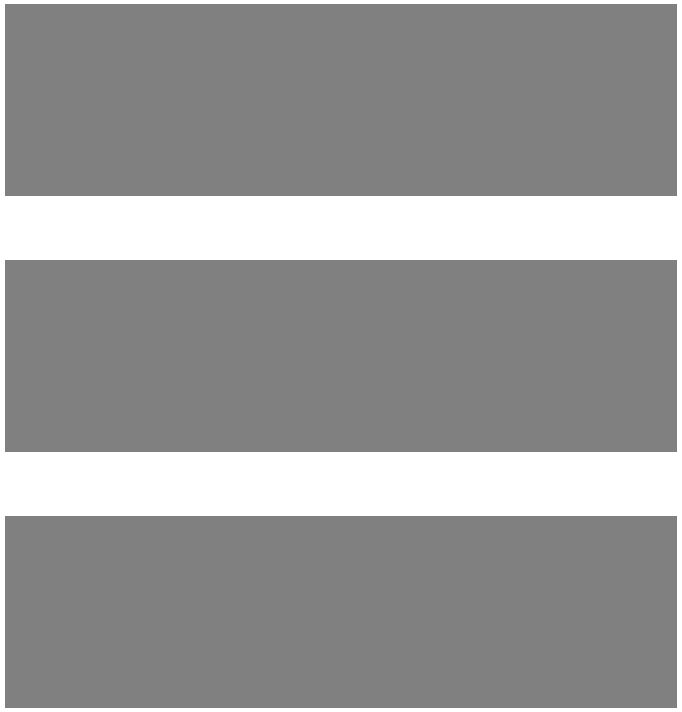
WHAT YOU SEE

white space - the counter

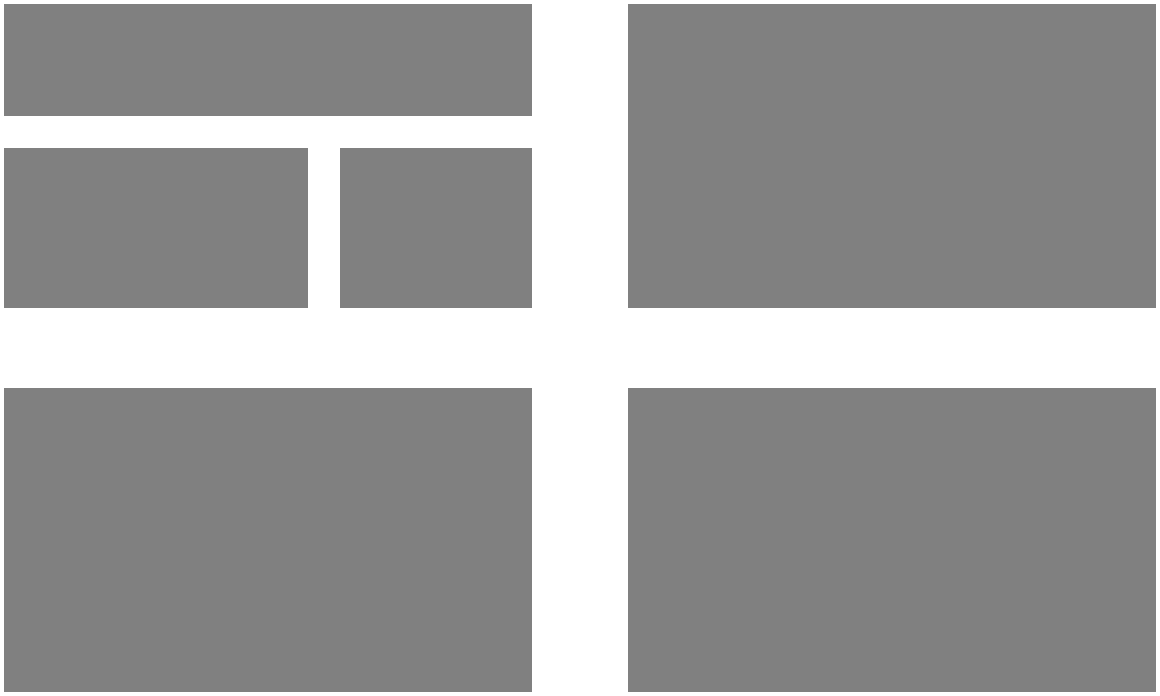
WHAT YOU SEE

THE GAPS BETWEEN

space to separate



space to structure

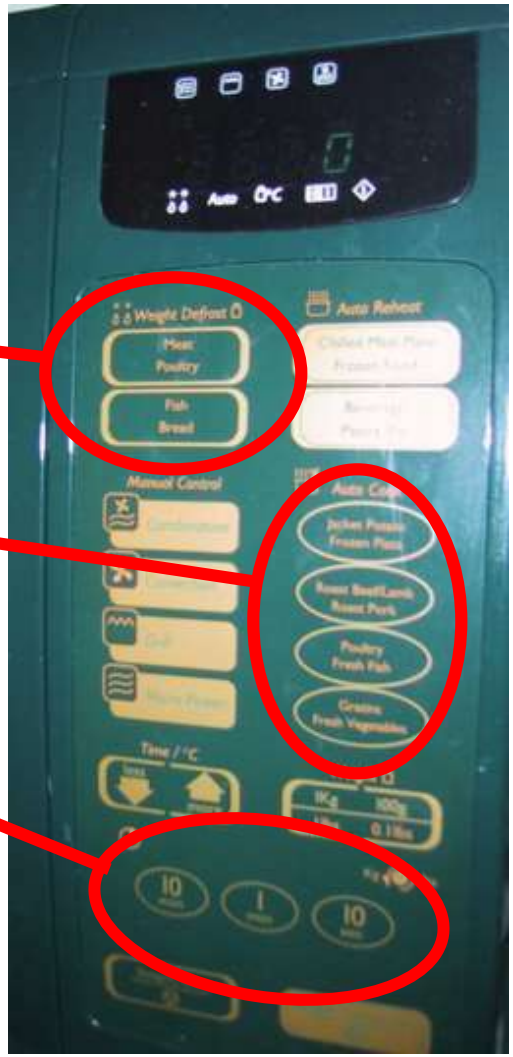


space to highlight



physical controls

- grouping of items
  - defrost settings
  - type of food
  - time to cook

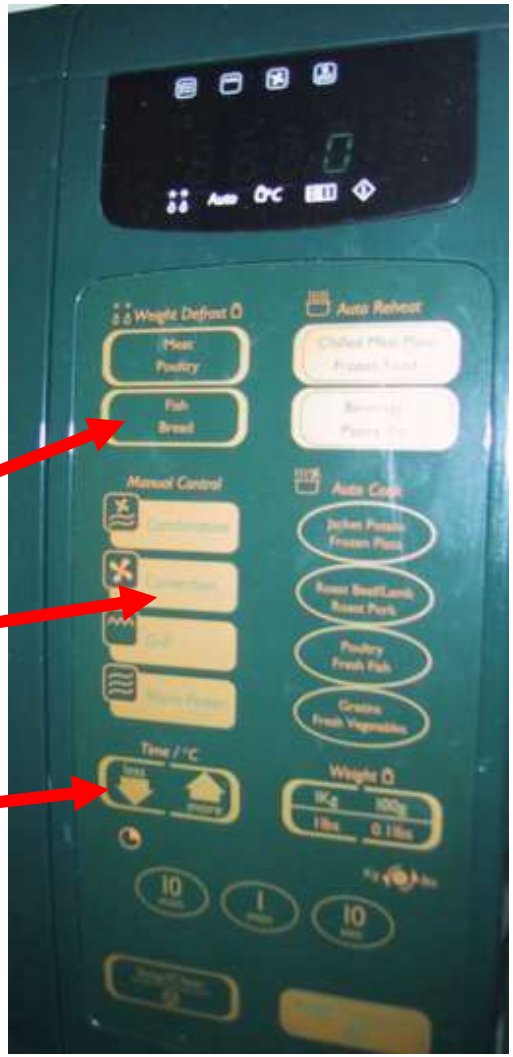


physical controls

- grouping of items
- order of items
- decoration

  different colours for different functions

  lines around related buttons (temp up/down)

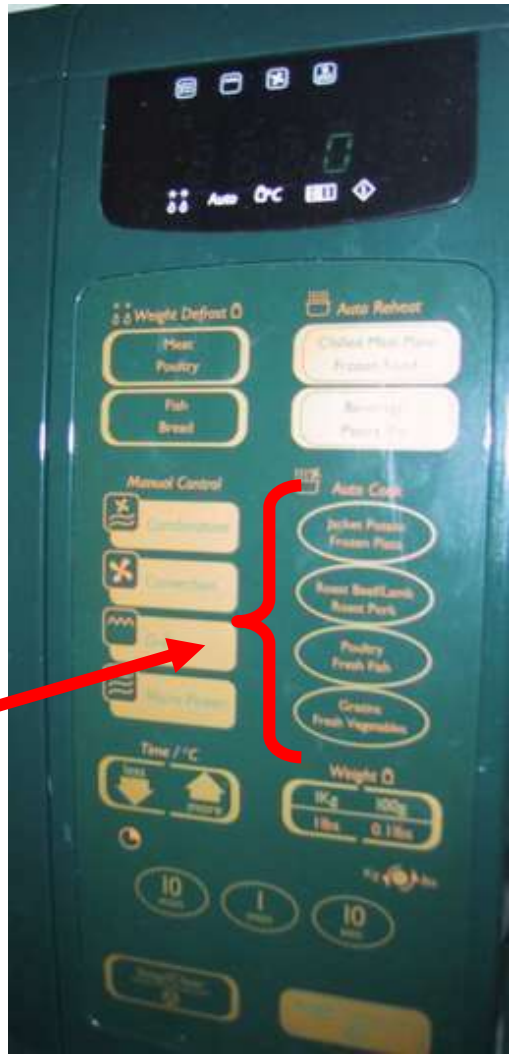


physical controls

- grouping of items
- order of items
- decoration
- alignment

centred text in buttons

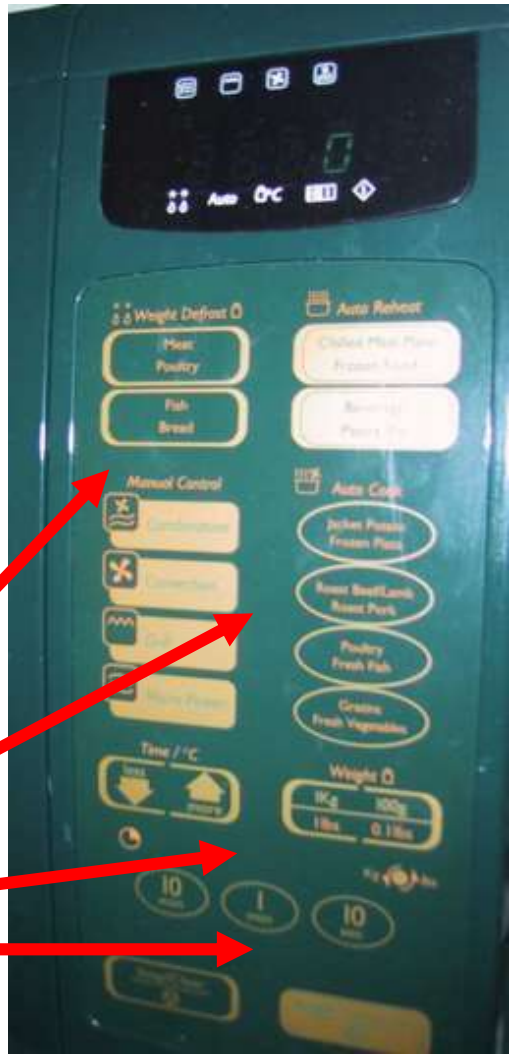
? easy to scan ?



physical controls

- grouping of items
- order of items
- decoration
- alignment
- white space

gaps to aid grouping

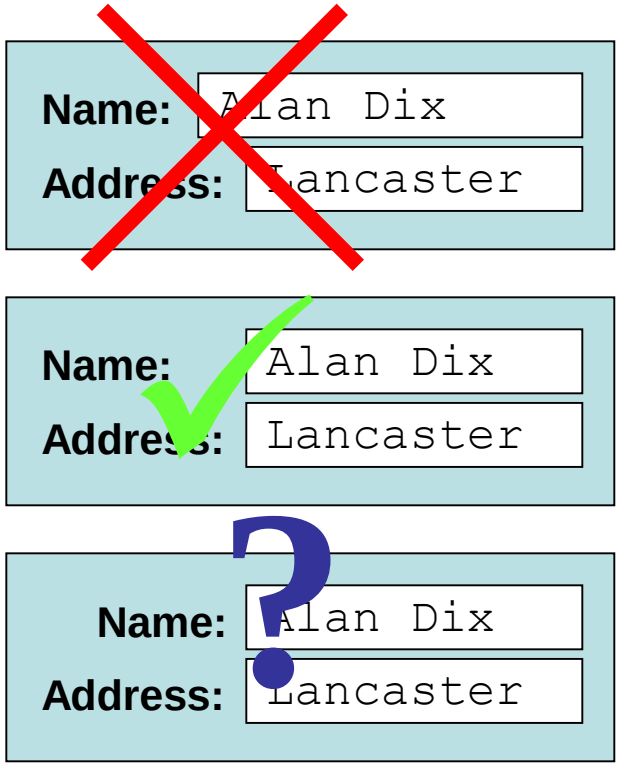


user action and control

entering information
knowing what to do
affordances

entering information

- forms, dialogue boxes
  - presentation + data input
  - similar layout issues
  - alignment - N.B. different label lengths



- logical layout
  - use task analysis (ch15)
  - groupings
  - natural order for entering information
    - top-bottom, left-right (depending on culture)
    - set tab order for keyboard entry

N.B. see extra slides for widget choice

appropriate appearance

presenting information

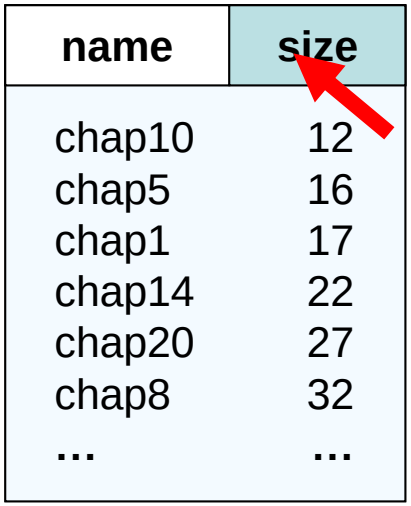
aesthetics and utility

colour and 3D

localisation & internationalisation

presenting information

- purpose matters
  - sort order (which column, numeric alphabetic)
  - text vs. diagram
  - scatter graph vs. histogram
- use paper presentation principles!
- but add interactivity
  - softens design choices
    - e.g. re-ordering columns
    - 'dancing histograms' (chap 21)

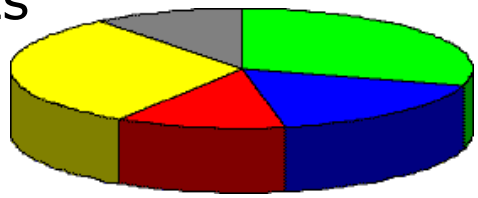


aesthetics and utility

- aesthetically pleasing designs
  - increase user satisfaction and improve productivity
- beauty and utility may conflict
  - mixed up visual styles ⇒ easy to distinguish
  - clean design – little differentiation ⇒ confusing
  - backgrounds behind text
    … good to look at, but hard to read
- but can work together
  - e.g. the design of the counter
  - in consumer products – key differentiator (e.g. iMac)

colour and 3D

- both often used very badly!
- colour
  - older monitors limited palette
  - colour over used because ‘it is there’
  - beware colour blind!
  - use sparingly to reinforce other information
- 3D effects
  - good for physical information and some graphs
  - but if over used …
    e.g. text in perspective!! 3D pie charts



bad use of colour

- over use - without very good reason (e.g. kids' site)
- colour blindness
- poor use of contrast
- do adjust your set!
  - adjust your monitor to greys only
  - can you still read your screen?

across countries and cultures

- localisation & internationalisation
  - changing interfaces for particular cultures/languages
- globalisation
  - try to choose symbols etc. that work everywhere
- simply change language?
  - use 'resource' database instead of literal text
    … but changes sizes, left-right order etc.
- deeper issues
  - cultural assumptions and values
  - meanings of symbols

    e.g tick and cross … +ve and -ve in some cultures
    … but … mean the same thing (mark this) in others



prototyping

iteration and prototyping

getting better …

… and starting well

prototyping

- you never get it right first time
- if at first you don't succeed …

